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Prof. Yonghui Hu
Editor, *Measurement*

Re: Revised submission of Ms. Ref. No. MEAS-D-26-10707
Gradient-Sensitive Functional Descriptors for Interferometric Surface Metrology: Complementing Amplitude-Averaging ISO Parameters

Dear Professor Hu,

Thank you for the opportunity to revise the manuscript. I am grateful to the three reviewers for their thorough and constructive comments, all of which have been addressed. The revised manuscript incorporates the following substantive changes.

In response to Reviewer 1: the definitional $R_a = L^1$ and $R_q = L^2$ equalities are now framed strictly as computational cross-checks and any suggestion of novelty in the functional-analytic relabelling has been removed; the four indicated references (Takeda et al. 1982; Kemao 2007; Feng et al. 2019; Galaktionov and Toporovsky 2026) have been added and are discussed individually in a new Introduction paragraph on fringe-analysis methodology; the redundancy of BV density and the $W^{1,2}$ seminorm as repeatability metrics is now stated explicitly, together with a map of which descriptors carry independent information; and a new paragraph acknowledges that the Flick flat analysis is an existence proof on a known certified feature, with a blind-detection protocol defined in Future Work.

In response to Reviewer 2: the state-of-the-art paragraphs have been rewritten so that every cited work is discussed separately with its key contribution; the former research-gap statement has been replaced by an explicit definition of the unresolved problem and of why existing approaches are insufficient; a new paragraph explains the acquisition-speed choices (rotation speed as the governing factor, 300–1200-fold angular oversampling of the metrological bandwidth, feasibility and untested risks of higher frame rates); the standards basis has been updated from ISO 4287 to ISO 21920-2:2021 throughout; the descriptor table now uses the (66.3 ± 4.7) nm format; and a new subsection explains rigorously why $R_z = RONt$ and $R_q = RONq$ hold by construction on the shared evaluation profile.

In response to Reviewer 4: a new subsection provides an indicative GUM-style type-B uncertainty budget quantifying laser wavelength stability, air refractive index, Abbe error/axis wobble, camera nonlinearity, and form-removal uncertainty, including a measured drift and noise-floor bound from an independent static stability session (-1.1 nm min^{-1}; 2.1 nm frame-to-frame), with the form-removal limaçon term identified as the leading bounded systematic; the discussion now contains a quantitative rule translating synthetic-defect responses to realistic inspection scenarios (bearing surfaces, optical components, precision seals) including the defect-to-background scaling that the reviewer highlighted; a standardised angular-unit convention (band-limit first, sample adequately, report in nm deg^{-1} and nm mm^{-1} with the declared filter cutoff) is defined in a new subsection with converted values for both artefacts and an empirical decimation demonstration (12-fold sampling change, $< 0.5\%$ change in band-limited descriptors); the Limitations and Future Work sections now state explicitly that a multi-session between-run campaign, wider artefact validation, and inter-laboratory comparison are prerequisites for any standardisation claim. Two further independent FN 111 sessions have been added as a between-run reproducibility assessment: for identical acquisition parameters every descriptor agrees within combined uncertainties (all $|E_n| < 1$; gradient descriptors within 4%), while a session at a changed rotation speed/frame rate shows 7–12% amplitude differences consistent with the form-removal bound, i.e. re-centring dominates over acquisition-rate changes. Additional new artefacts could not be acquired within the revision period; the claims of the paper have been scoped accordingly and these steps are defined as a concrete validation programme.

A detailed point-by-point response to every comment accompanies this letter (`response_to_reviewers.pdf`), together with a marked version of the manuscript (`banach-elsevier-marked.pdf`) highlighting all changes.

I hope the revised manuscript is now suitable for publication in *Measurement*.

Yours sincerely,

Dawid Kucharski